

ADITYA CHUNDURI

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EDUCATION

University of Southern California

M.S., Applied Data Science

Los Angeles, CA

Jan 2024 – Dec 2025

SRM Institute of Science and Technology

B.Tech., Computer Science & Engineering, AI/ML Specialization

Chennai, India

Jun 2019 – Jul 2023

TECHNICAL SKILLS

Languages & Engineering: Python, SQL, JS/TypeScript, Bash, C/C++ ▪ FastAPI, React/TypeScript, Docker, pytest, GitHub Actions, SQLite, Neo4j, ChromaDB, Ollama

ML / CV / Audio: PyTorch, TF/Keras, scikit-learn, OpenCV, U-Net, YOLO, Diffusers, librosa, Demucs, Beat This!, CLAP, Essentia

LLM Security & Eval: Inspect AI, AgentDojo, OWASP LLM, OWASP Agentic, MITRE ATLAS, AIVSS ▪ prompt injection, MCP tool poisoning, RAG poisoning, red-teaming

PROJECTS

Agent Shield (Python, Inspect AI, AgentDojo, ChromaDB, pytest, GitHub Actions)

[GitHub](#) | [preprint](#)

- Authored an open-source LLM agent evaluation framework on Inspect AI covering 6 live attack modules and 24 attack IDs (prompt injection, MCP tool poisoning, RAG/memory poisoning, exfiltration, social engineering, behavioral drift); introduced Transparency Rate alongside Benign Utility, Utility Under Attack, and Targeted ASR in a six-cell outcome model extending AgentDojo's 2x2 matrix.
- Ran seed-pinned evaluations across 4 models with Wilson 95% CIs at $n=20$ — three of four score $TR=0.000$ on anchored injection; spotlighting raises TR to 0.800 but adds 0.200 to ASR via paraphrase residue. Mapped all attacks to OWASP LLM, OWASP Agentic, MITRE ATLAS, and AIVSS; released as a Zenodo preprint with framework, seeds, and logs.

AI RemixMate (FastAPI, React/TypeScript, Demucs, Beat This!, CLAP, librosa, Essentia, pyloudnorm)

[GitHub](#)

- Built a full-stack DJ engine: FastAPI async job queue with SQLite persistence, React/TypeScript frontend (8 pages, SSE live streaming, Zustand state), and -14 LUFS mastering pipeline — 374+ tests across unit, behavioral, and integration suites, GitHub Actions CI, 9+ organic stars.
- Implemented a research-grade MIR core — TIV harmonic scoring (Bernardes et al. 2016), Beat This! (ISMIR 2024) downbeat detection with bar-grid snapping, cosine-taper stem bass ramp, FxNorm per-stem LUFS normalization — plus CLAP 512-D semantic search, Essentia energy arc modeling, and rekordbox XML + Serato GEOB cue export.

AkashicTree (Ollama, Diffusers, Replicate FLUX.1, ElevenLabs, Coqui TTS, FFmpeg)

[GitHub](#)

- Built a model-agnostic multimodal pipeline converting topic prompts into reviewed scripts, generated images, narrated audio, and FFmpeg-assembled vertical videos, with tiered local and cloud backends (Ollama, Diffusers, FLUX.1, ElevenLabs).

AI Health Journal (Flask, Ollama, ChromaDB)

[GitHub](#)

- Built a fully local Flask/Ollama RAG assistant with ChromaDB retrieval and session memory; compared dense, BM25, RRF hybrid, and cross-encoder reranked retrieval on a provisional gold set — MiniLM hybrid+rerank $nDCG \sim 0.78$ at 504 ms versus hashing hybrid at ~ 0.8 ms.

WORK EXPERIENCE

USC Viterbi School of Engineering

Los Angeles, CA

Research Assistant, Computer Vision & Medical Imaging

Aug 2024 – Dec 2024

- Engineered a TensorFlow/Keras U-Net experimentation pipeline for retinal artery-vein segmentation on DRIVE and CHASE_DB1 fundus datasets, with per-run AUC logging from ~ 0.93 to 0.97 across model variants.
- Integrated rotation, flip, and contrast augmentation into the preprocessing pipeline (raising validation AUC by ~ 0.04 over baseline); built reusable model-comparison workflows with Keras history and TensorBoard artifacts for faculty review.

SSN College of Engineering

Remote

Research Intern, Computer Vision & Object Detection

Jun 2021 – Jul 2021

- Built YOLO-based object-tracking prototypes and reusable PyTorch/TensorFlow modules; annotated a 1,200+ frame video dataset and benchmarked YOLOv5 vs. YOLOv4 using $mAP@0.5$ on a held-out test set.

PUBLICATIONS & CERTIFICATIONS

Chunduri A., "Beyond Attack Success Rate: Measuring Operator-Facing Transparency in LLM Agent Security," *Zenodo* preprint, 2026.

[doi:10.5281/zenodo.20789431](https://doi.org/10.5281/zenodo.20789431)

Certifications: IBM Data Science & AI Professional Certificates | Machine Learning — Stanford University (Coursera) | Computer Vision & Image Processing — IBM (Coursera)